Sprawozdanie 3

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```
## Start:  AIC=-66.25
## Y ~ X1 + X2 + X3 + X4 + X5 + X6 + X7
##
##        Df Sum of Sq    RSS     AIC
## - X5    1         1    117  -67.46
## <none>                 117  -66.25
## - X1    1       731    847  294.46
## - X7    1      1195   1312  374.49
## - X3    1      6844   6960  679.85
## - X4    1      8288   8404  714.35
## - X2    1     63879  63996 1085.85
## - X6    1    121923 122040 1203.98
##
## Step:  AIC=-67.46
## Y ~ X1 + X2 + X3 + X4 + X6 + X7
##
##        Df Sum of Sq    RSS     AIC
## <none>                 117  -67.46
## + X5    1         1    117  -66.25
## - X1    1       743    861  295.29
## - X7    1      1201   1318  373.30
## - X3    1      6869   6986  678.53
## - X4    1      8291   8409  712.44
## - X2    1     64136  64254 1084.58
## - X6    1    123690 123807 1204.61
```

```
summary(model)
```

```
##
## Call:
## lm(formula = Y ~ X1 + X2 + X3 + X4 + X6 + X7, data = dane)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.53673 -0.64916  0.06226  0.57655  1.79445
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.26351    0.28161  -0.936    0.351
## X1           2.02275    0.06056  33.401   <2e-16 ***
## X2           4.00096    0.01290 310.270   <2e-16 ***
## X3           6.24122    0.06147 101.540   <2e-16 ***
## X4           4.00864    0.03593 111.557   <2e-16 ***
## X6           5.98123    0.01388 430.878   <2e-16 ***
## X7           1.03655    0.02442  42.452   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8162 on 176 degrees of freedom
```

```
## Multiple R-squared:  0.9995, Adjusted R-squared:  0.9995
## F-statistic: 5.69e+04 on 6 and 176 DF,  p-value: < 2.2e-16
```

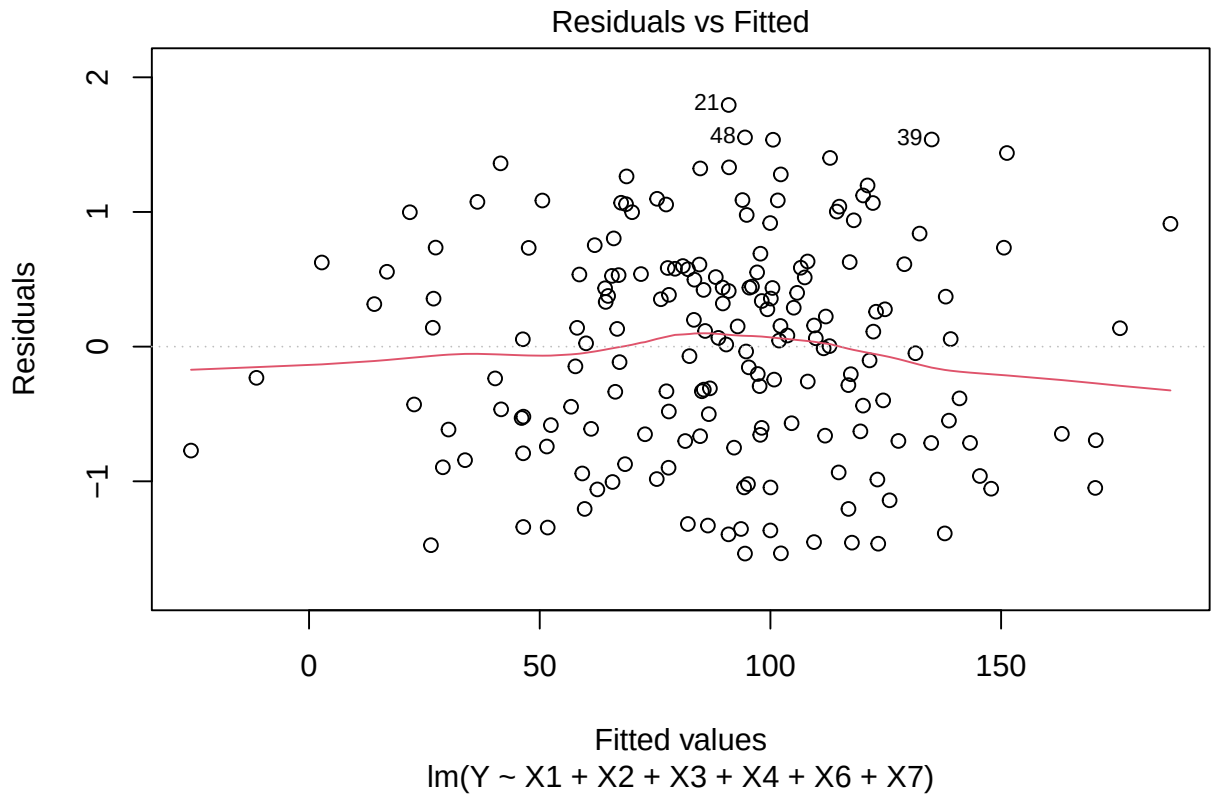
1.1 Estimate Std 1.2. F-statistic: p-value: ¡ 2.2e-16 1.3 z czwartej kolumny wszystkie zmienne istotne

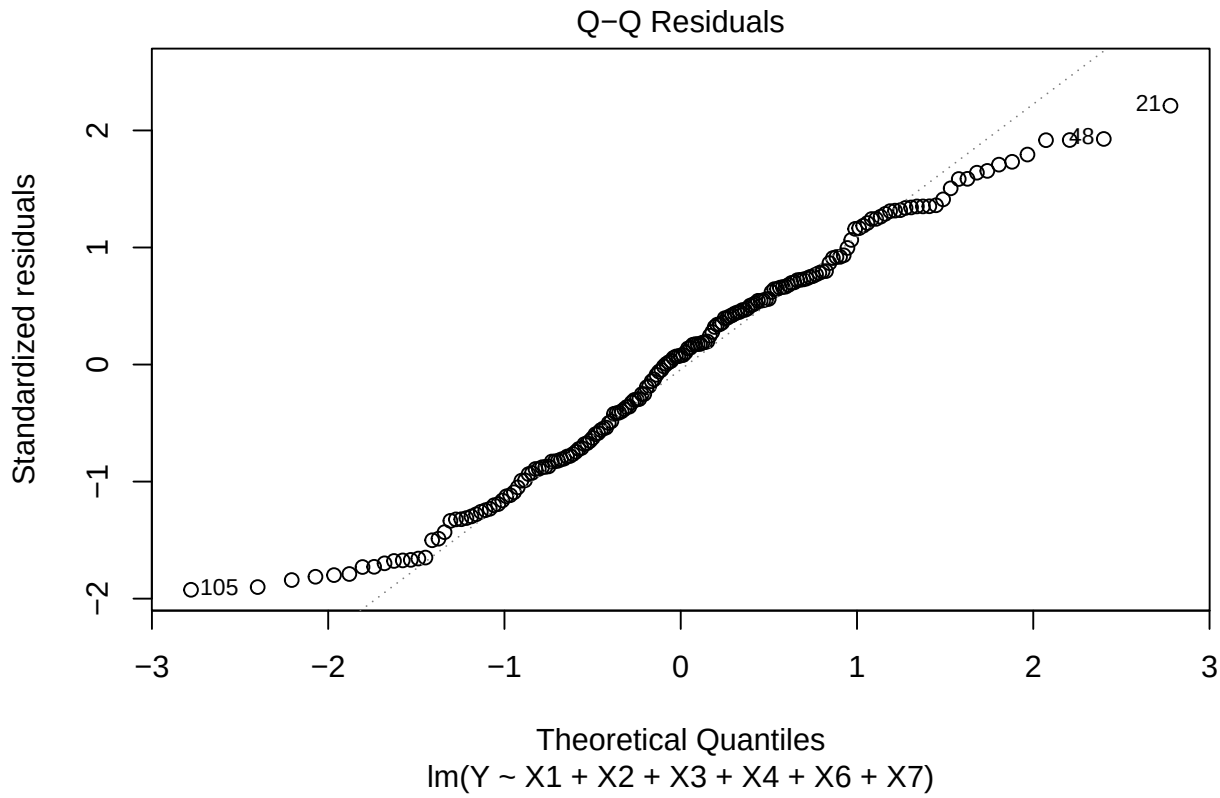
```
confint(model, level=0.95)
```

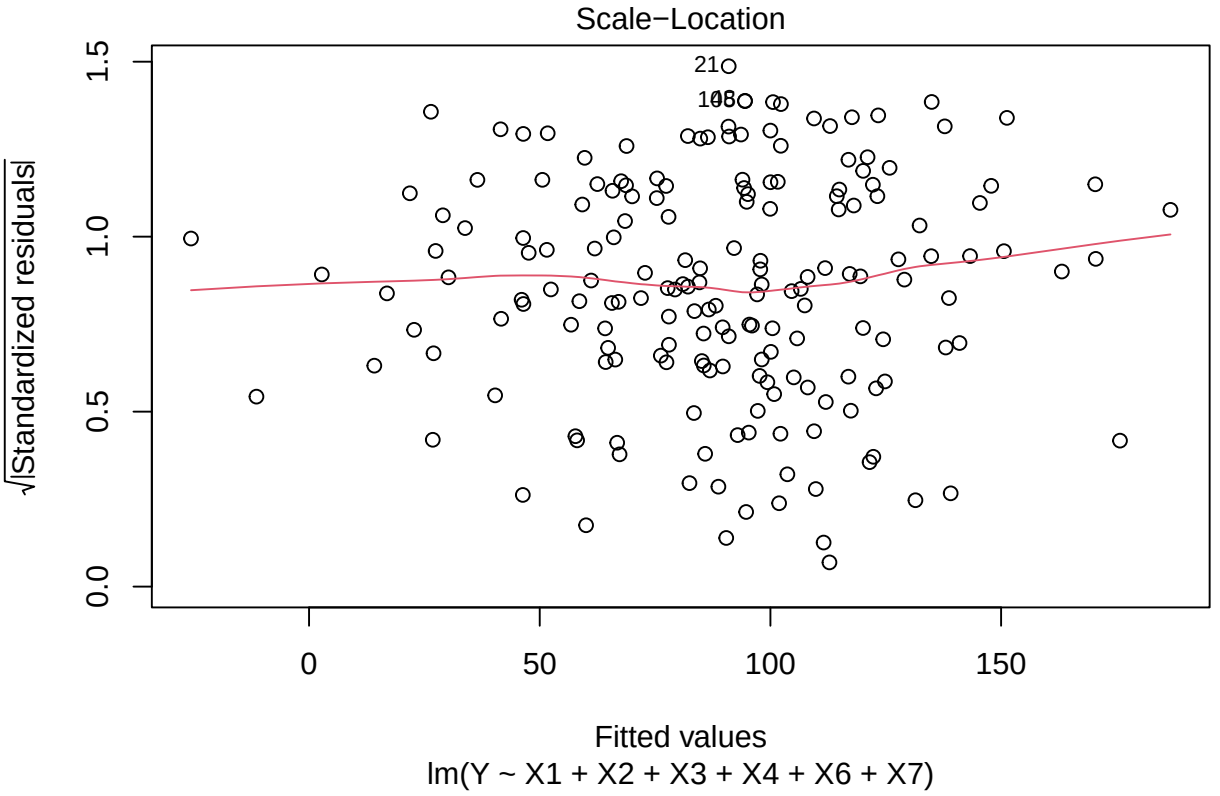
```
##                 2.5 %    97.5 %
## (Intercept) -0.8192737 0.2922544
## X1           1.9032348 2.1422695
## X2           3.9755064 4.0264041
## X3           6.1199129 6.3625215
## X4           3.9377217 4.0795534
## X6           5.9538381 6.0086294
## X7           0.9883591 1.0847335
```

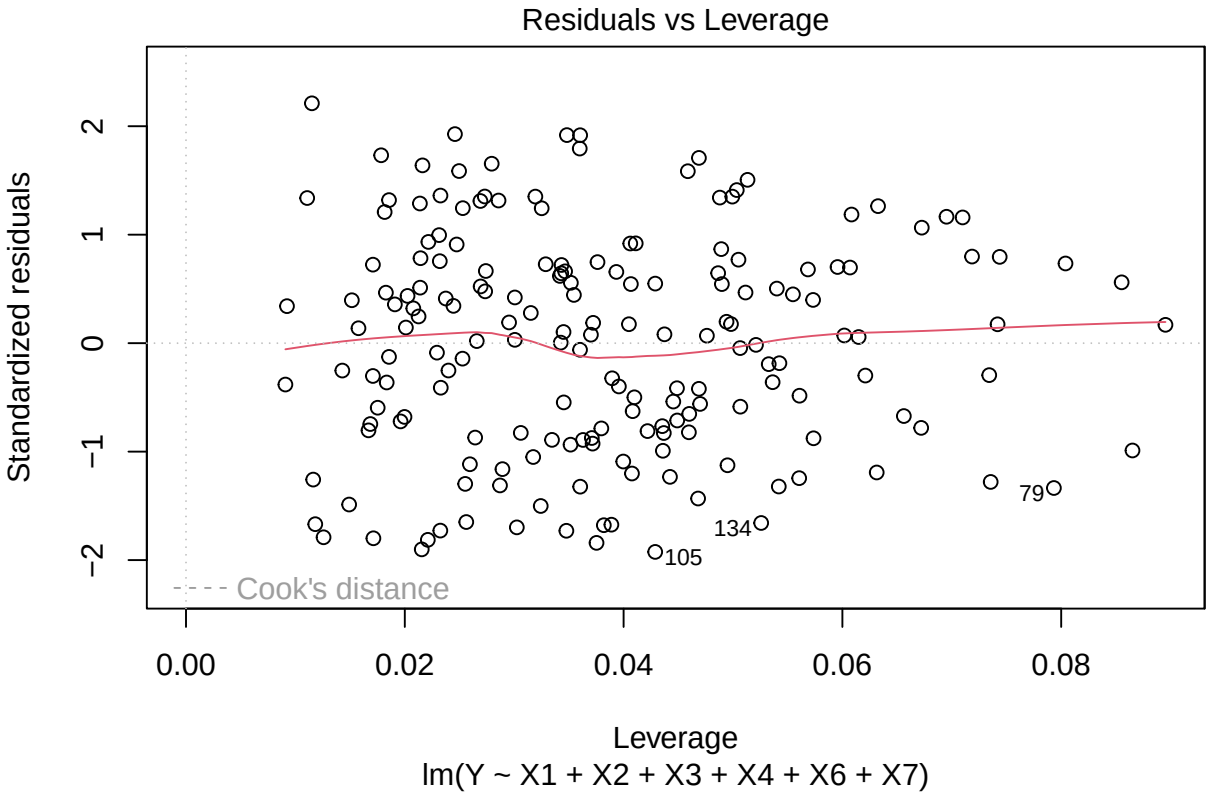
przedziały ufności nie zawierja zera Adjusted R-squared: 0.9995 SUPER model dopasowuje sie do danych

```
plot(model)
```




Q−Q Residuals
Standardized residuals
Theoretical Quantiles
lm(Y ~ X1 + X2 + X3 + X4 + X6 + X7)
105
48
21

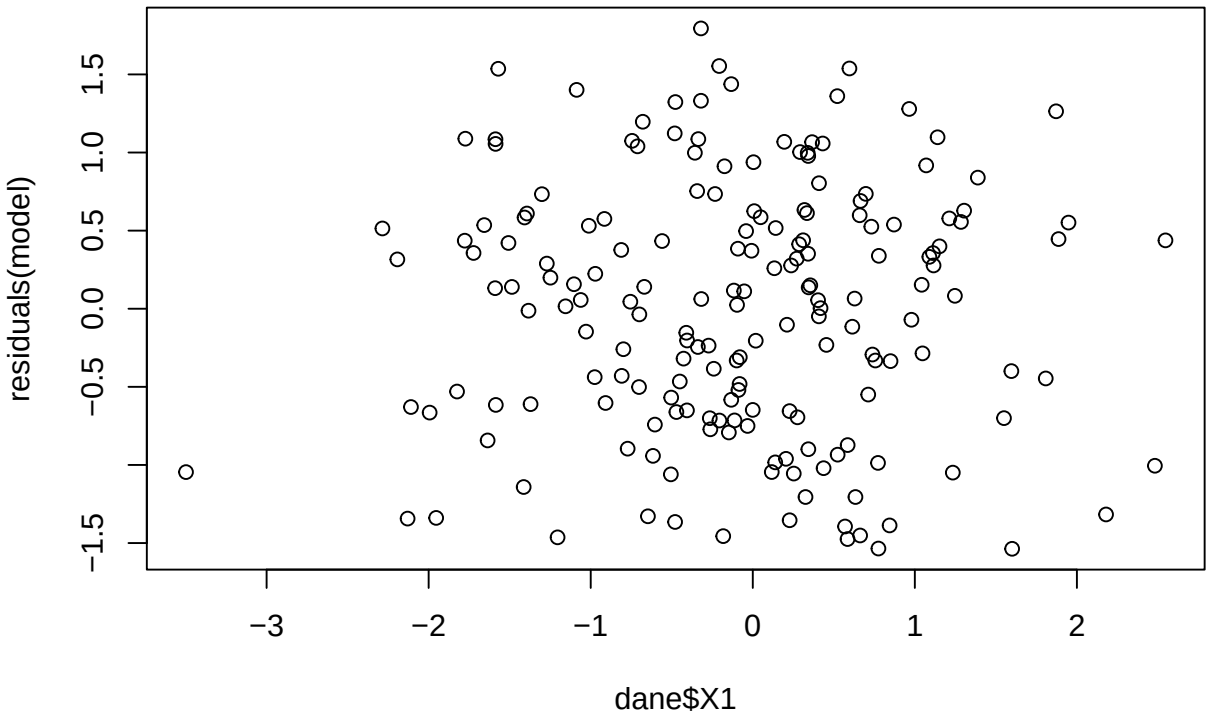

Scale−Location
√|Standardized residuals|
Fitted values
lm(Y ~ X1 + X2 + X3 + X4 + X6 + X7)
21



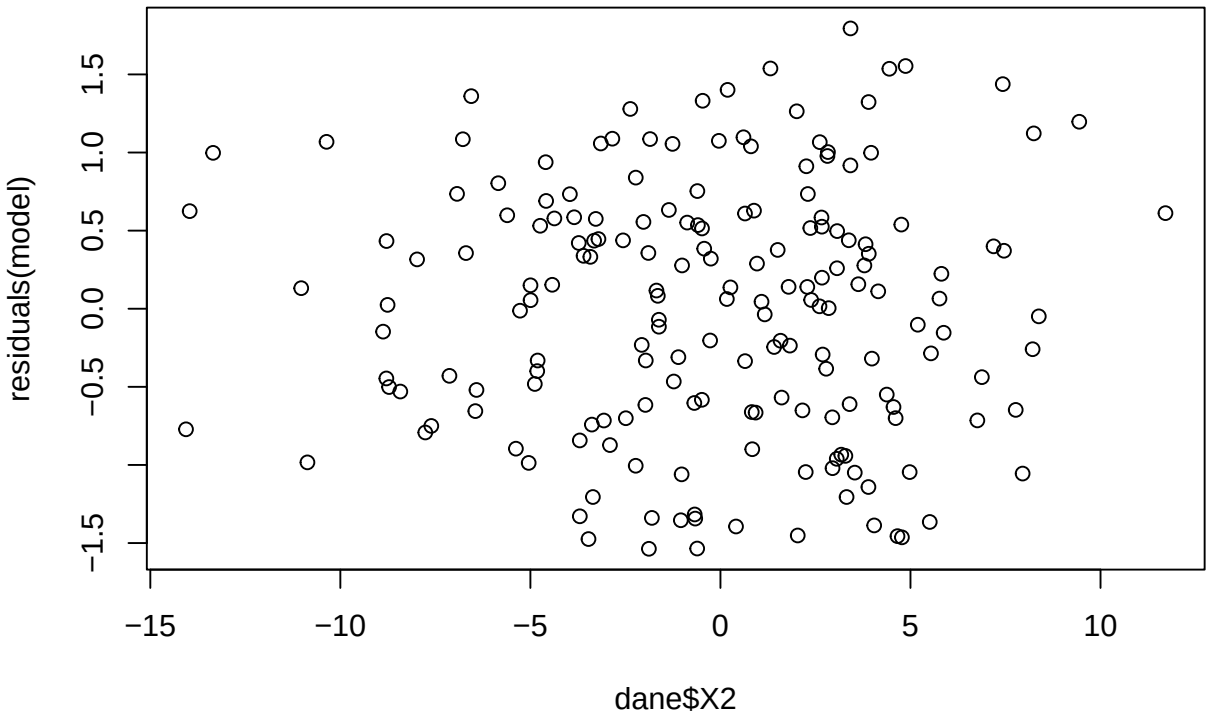
Wykres Q-Q wygląda w miare dobrze pojawiają się odstępstwa na ogonach

Residuals vs Fitted wygląda dobrze linia jest stosunkowo prosta w okolicach zera

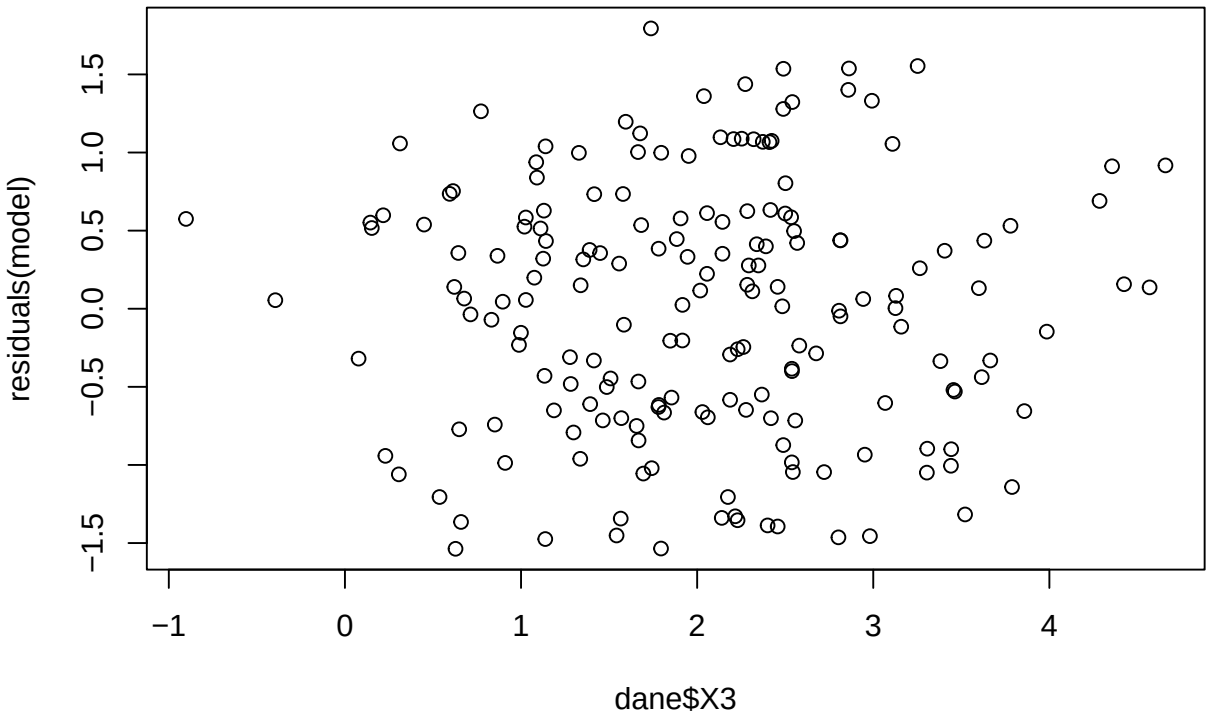
```
plot(dane$X1, residuals(model))
```



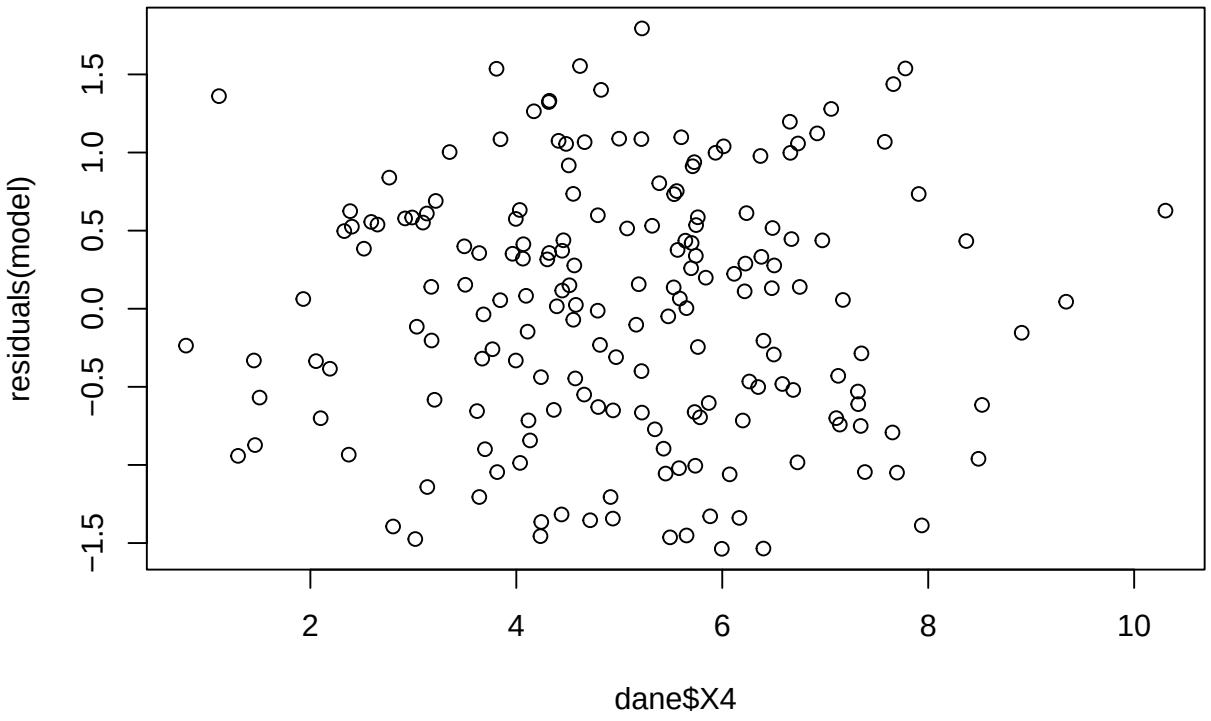

```
plot(dane$X2, residuals(model))
```



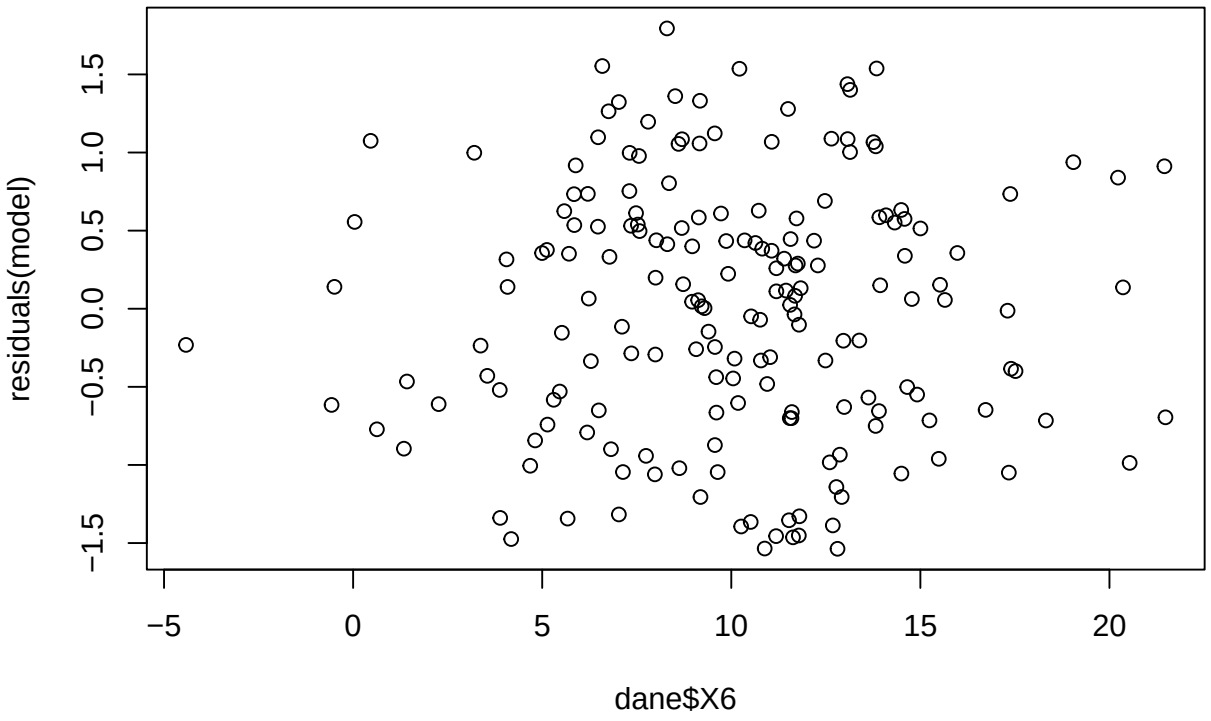

```
plot(dane$X3, residuals(model))
```



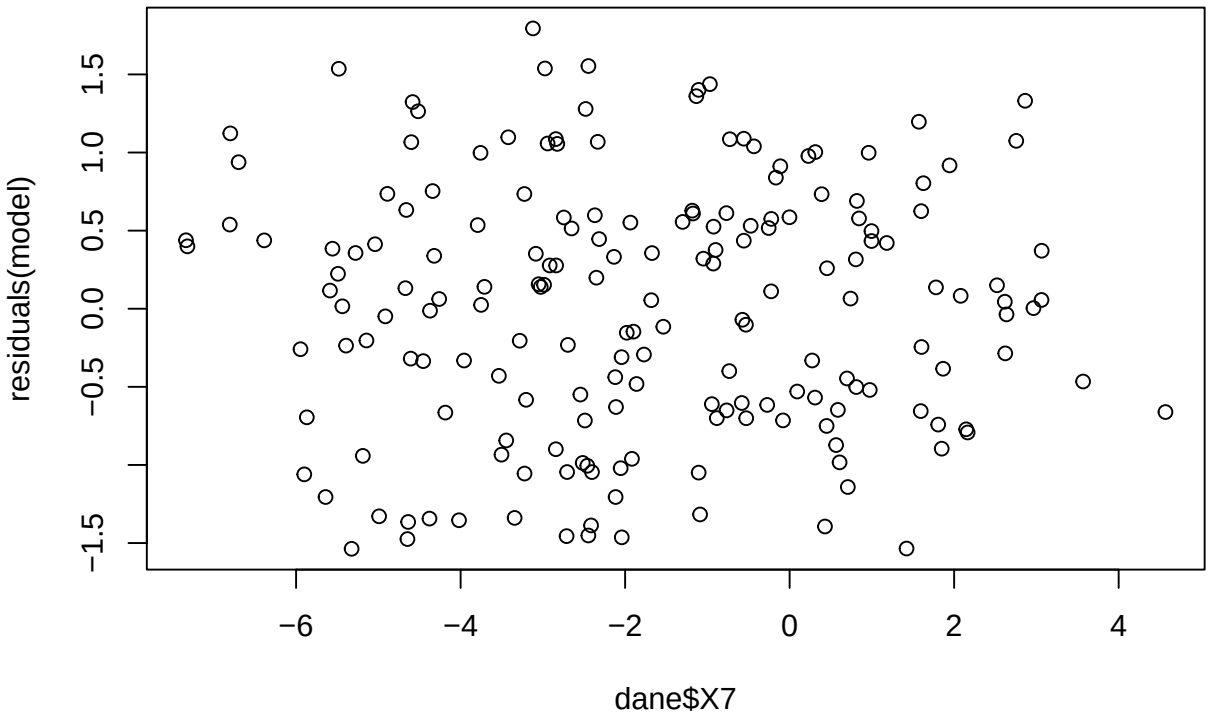

```
plot(dane$X4, residuals(model))
```




```
plot(dane$X6, residuals(model))
```




```
plot(dane$X7, residuals(model))
```



Wykresy reszt wyglądaja ok tzn błędy wyglądaja na losowe